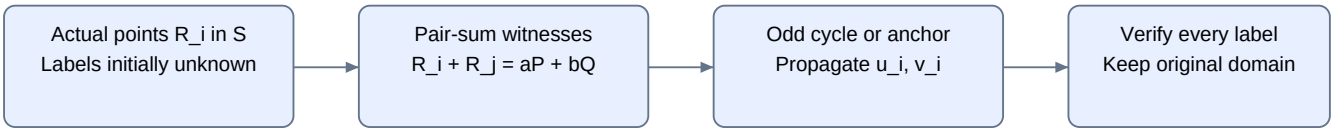
Two next leads for known-label construction

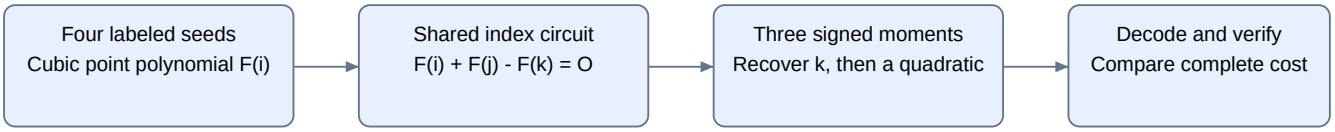
Lead 1: fixed domain, labels recovered from verified relations



Triangle identity: 2 R_1 = T_12 - T_23 + T_31. Cycle label defects can solve the DLP.

Pair-table build, failed queries, graph work and group verification are charged.

Lead 2: four-seed cubic generator with one-minus signed decoding



Proposals, not measured speedups. Scalar field F_r and coordinate field F_p stay distinct.

2026-10-28. **Research proposals with derived algebra and correctness controls; no attack improvement or new timing result is claimed.** The original goal remains an efficient construction into a small usable coordinate domain, with lower complete cost. Repeating its blocker is not the next research step.

This note develops two alternatives to the earlier fixed-norm/cubic lifts. It does not change PR #1215, its historical archive, or official hypothesis status. No new elliptic-curve experiment or performance run was launched. The setting is a short Weierstrass curve over F_p with p>3, a prime-order subgroup G=<P> of odd order r>3, and Q in G. Parameters are symbolic; this note registers no new curve/candidate ID or measured run ID.

1. Obtain individual labels from pair relations

Choose the coordinate domain first, enumerate its actual subgroup-usable curve points, and fix one orientation R_i per usable abscissa. The points initially have unknown representations. A verified decomposition

```
R_i + R_j = T_e = [a_e]P + [b_e]Q
```

is an edge in a label graph. It supplies a linear equation for the two unknown labels, rather than requiring either label in advance. For a triangle:

```
2 R_1 = T_12 - T_23 + T_31.
```

Thus R_1 has label ((a_12-a_23+a_31)/2, (b_12-b_23+b_31)/2), modulo the odd prime group order r. The other two labels follow. More generally, a connected non-bipartite component determines labels from any odd cycle; a bipartite component needs one labeled anchor. If different cycles yield inconsistent coefficient pairs, their discrepancy is a zero relation [a]P+[b]Q=O. For b != 0 it yields a candidate DLP solution, which must be independently verified and counted as an early success.

This directly attacks the previously unresolved u_i,v_i construction. The domain need not move, and its original sparse membership equation is retained; only actual curve points become graph vertices. The technique is related to classical relation linear algebra and large-prime graphs, not a claim that graph methods are new. B denotes usable abscissae, not raw ambient-domain size. Charge lift and subgroup filtering. An ambient binomial may be retained with curve/subgroup and exact chosen-base checks; it is not the filtered set's two-term vanishing polynomial. The sparse ambient equation alone never certifies a labeled vertex. The atlas already tracks rank-aware large-prime graphs in LEAD-ICE-relation-graph-rank-large-prime. The proposed variant uses pair equations to create labels inside the chosen domain, not only to cancel outside large primes from an existing relation pool.

Complete first implementation

1. Enumerate and verify the B usable vertices. Build a dictionary of all distinct pair-sum targets with their full edge lists; do not discard pair collisions. This costs O(B^2) group additions and storage in this control. 2. Generate labeled T=Q+[u]P with the existing charged window sampler. Look up every target, record failed queries, and insert every verified matching edge. 3. Propagate labels with `label_graph.py`. Stop at the declared sufficient labeling/rank criterion or an independently verified earlier DLP solution. 4. Verify recovered labels in the curve group, then use the existing chosen-base relation solver. Do not report the graph update cost as the whole constructor.

If K distinct targets are indexed, a uniform query hits the dictionary with probability exactly K/r. Graph connectivity and edge distribution must be measured, not assumed. Under the explicit additional random-edge model with K=Theta(B^2), collecting O(B log B) edges suggests O(r log B/B) queries and an O~(B^2+r/B) construction scale. This is a heuristic cost model, not an established bound for multiplicative-coset points and not a rho improvement. The optimized interval/multi-target BSGS constructor is a mandatory control.

The first discriminating question is whether this complete constructor beats the best existing one on the same saved point domain. If not, keep its pair table as a control and investigate a compressed pair-decomposition oracle; do not recycle an old coset as a new mechanism or claim the O(B^2) table is free.

2. A four-seed cubic base with signed-moment reconstruction

Let r>3 be prime and I={0,...,B-1} with B<=r. Sample four independent uniform label pairs (a_j,b_j), and construct T_j=[a_j]P+[b_j]Q. Define

```
F(i) = T_0 + [i]T_1 + [i^2]T_2 + [i^3]T_3.
```

Every point is labeled by the same scalar polynomials in a_j and b_j. Forward differences require three group additions per index after charged seed/initialization work. The previous degree-five construction required five; this comparison is an operation count for that generator alone. An optimized independent window sampler is already cheaper per point and remains a control.

Keep the signed events F(i)+F(j)-F(k)=O, with three distinct indices and the positive pair unordered. Put s_m=i^m+j^m-k^m for m=1,2,3 and

```
D = (s_1^2-s_2)/2 = (i-k)(j-k).
k = (-s_1^3+3 s_1 s_2-2 s_3)/(6D).
i+j = s_1+k,       i*j = D+k s_1.
```

D is nonzero for distinct indices. The negative index and the unordered positive pair are therefore uniquely determined by the three moments; the pair is recovered from a quadratic. Index bounds, discriminant and the actual point relation still have to be checked. This is a Prony/Padé-style signed-moment reconstruction, not a novel general reconstruction theorem.

Written probability argument, scoped to one-minus triples

Each event has coefficient vector (1,s_1,s_2,s_3) on four uniform seed points. The reconstruction above makes distinct event vectors distinct and hence nonproportional. Their zero-sum events are pairwise independent. Conditional on the group relation, the Q-label coefficient is uniform. Thus useful-event probability is q=(r-1)/r^2, and useful-event indicators are pairwise independent too, by applying the rank-two coefficient map to independent seed label pairs.

For M=3*binomial(B,3), the second-moment bound gives

```
Pr(at least one useful event) >= M*q/(1+(M-1)*q) - B^2/r.
```

The final subtraction excludes identity vertices and F(i)=+/-F(j), by the pairwise-independent evaluation bound. Truncate a negative bound at zero. For B asymptotic to r^(1/3), this gives 1/3-o(1). This is a derived written bound, not a measured relation rate or a checked Lean probability theorem.

Do not assume the all-plus events are independent of this family. There is an explicit counterexample modulo 11: the moment vector of (0,1,2), all plus, equals three times that of (3,10; negative 1). Restricting to one-minus events keeps 3 of the 4 global-sign classes, a transparent 4/3 event-count loss.

Dedicated-solver opportunity

Compile the cubic Horner generator using shared index bits and complete curve arithmetic; use signed-moment reconstruction as a propagation/decoding rule. The scalar field F_r and coordinate field F_p remain distinct. Compare this with the previous degree-five circuit, independent window sampling, direct pair search and the existing trace-fiber solver on matched inputs. Enlarging the base by (4/3)^(1/3) equalizes the leading signed-event count, but does not establish a timing improvement. Full-width moment scalar multiplication may cost more than direct small-index Horner evaluation and must be charged.

This lead changes the domain to the explicitly generated set of B abscissae; it does not assert a small multiplicative coordinate product set. Its possible advantage is a more structured generator circuit and exact signed decoding. Small circuit size alone does not imply efficient solving. A useful solver must reduce complete cost against the optimized pair baseline, not merely remove variables or display low-degree equations.

Validation and next decision

`test_algebra.py` checks symbolic identities and abstract-group graph controls, including a cycle that exposes a logarithm. These are correctness controls; they contain no empirical elliptic-curve performance comparison. Run:

```
python -m unittest discover -s docs/known-label-next-leads-20261028 -p test_algebra.py -v
```

`Algebra.lean` contains proposed ring-identity proofs. Lean and Lake were unavailable on this host, so that source is UNBUILT and unverified. The full graph-to-elliptic-group bridge, finite-field signature injectivity, and probability statement remain unformalized. Symbolic checks do not replace them.

Before any new benchmark, freeze the protocol and full cost boundaries, obtain the repository checkpoint receipt, and follow current curve-identity, execution, isolation and matched-baseline rules. No experiment is scheduled by this note. Prioritize the graph constructor; the second test is the cubic circuit with the one-minus decoder. Preserve failures, memory and scalar replay costs.

Sources and novelty scope

- Existing code/data and exact unresolved requirement: Frobenius/block publication, including its later accounting correction. No historical archive is edited. - The program atlas was consulted and matched; see `atlas-matches.json` and its pinned source revision. Graph algebra, relation linear algebra, polynomial sampling and Prony reconstruction are established tools. These are concrete next variants for this prototype, not claims of global novelty. Its 147 lead files and 52 known-result files were downloaded at a frozen revision and Git-blob hashes checked. The first matcher invocation failed because the snapshot root was absent from PYTHONPATH; that log is retained. The successful invocation used the same unchanged matcher and catalog. - Petit, Kosters, Messeng, PKC 2016: sparse/composed coordinate domains already occur in prior work. A better complete construction/solver cost is still owed. - Diem, double-large-prime index calculus: related graph methods for curves; its genus>=2 result is not a result for this prime-field elliptic-curve proposal.

No new curve, isogeny or verified map is asserted. The existing volcano-IC graph set and curve-catalog roles were checked for applicability; no canonical edge or curve record changes. The diagram in this note is algorithm/data flow only.